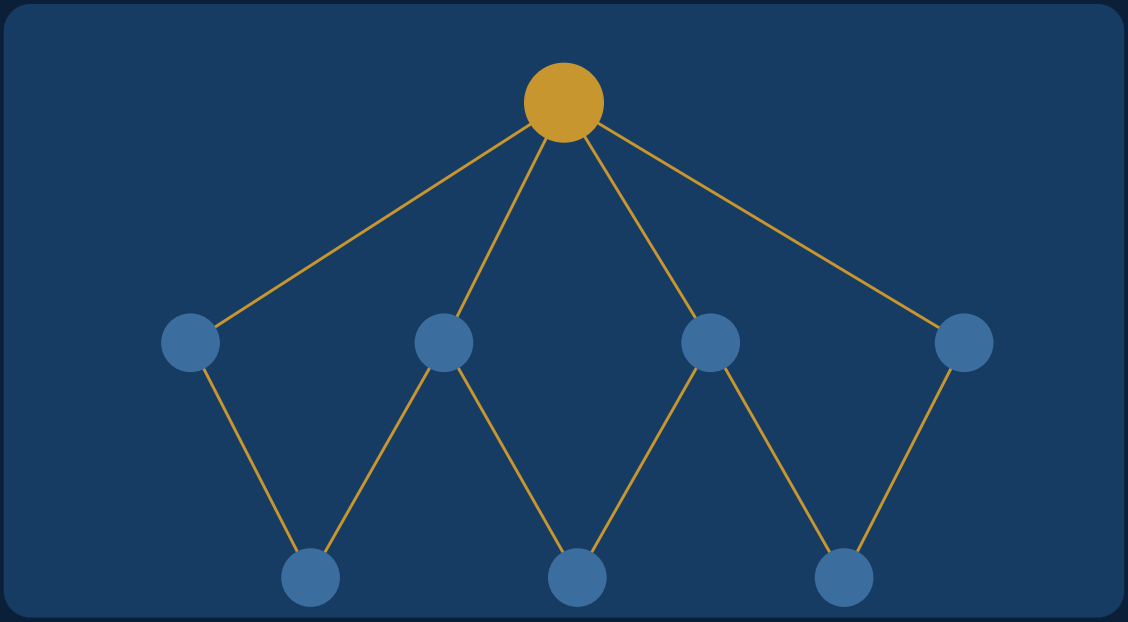
DBMS STUDY SERIES

For OAVS (TGT Computer Science), KVS, NVS, DSSSB, OSSSC and similar competitive examinations

Chapter 4
Network Database

4.1 Introduction



Records connected through owner–member set relationships

A complete, exam-oriented introduction to the CODASYL / DBTG Network Data Model

Definitions • History • Structure • Diagrams • Advantages & Limitations • Quick Revision

1. Definition & Background

A **Network Database** organises records as an **interconnected network** rather than a strict tree — a record can be related to more than one other record. The **Network Database Model** represents data through **record types** linked by **set types**, where a member record may have **more than one owner**, unlike the hierarchical (tree) model.

■ EXAM POINT — History — in simple words

The network model was made in the **1960s** to fix a problem with the tree (hierarchical) model — it could not show one record linked to **many** owners. **Charles Bachman** built the first such system, called **IDS**, at General Electric. Later, a group called **CODASYL** set up a team named **DBTG** (Data Base Task Group). This team wrote official rules for the model in **1969**, updated in **1971**. That is why the model is also called the **CODASYL model** or the **DBTG model**. A well-known software that used this model was **IDMS**.

2. Key Terminology

Term	Definition
Record	One occurrence of an entity — a set of related data item values.
Record Type	The template/definition for a group of similar records.
Data Item	The smallest named unit (field) of data within a record.
Set Type	A named 1:M relationship between one owner and one member record type.
Owner	The record on the “one” side of a set.
Member	The record on the “many” side of a set; can belong to more than one set.
Set Occurrence	One actual instance of a set type for a specific owner.
Link	The pointer physically connecting an owner record to its member records.

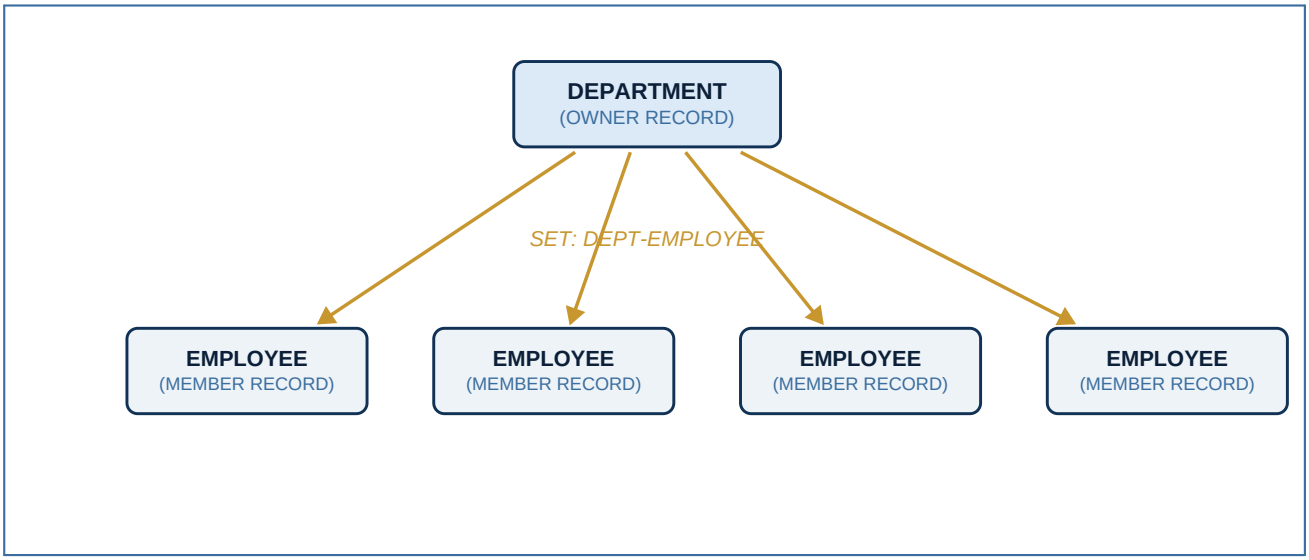
■ IMPORTANT TERM — Rule to remember

A set has exactly **one owner** type, but a **member** type may belong to more than one set — this is what allows many-to-many relationships.

3. Structure at a Glance

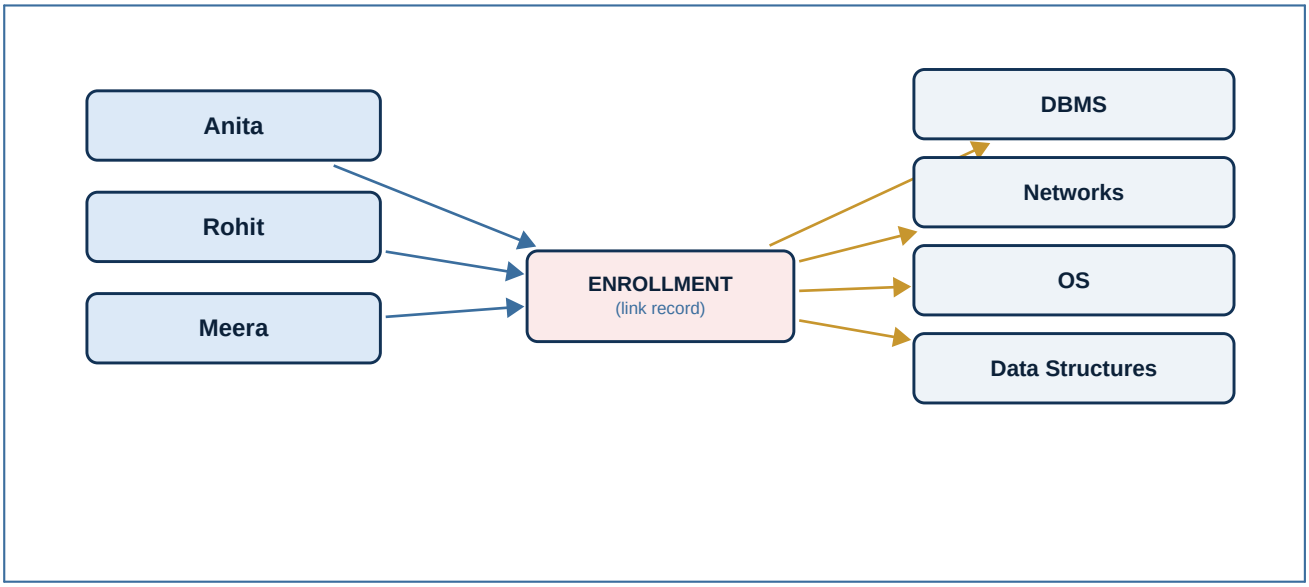
Every network database is built from **record types** connected by **set types**. Recognise these two situations instantly: a simple owner–member set (Figure 1), and a many-to-many relationship built from two sets sharing one link record (Figure 2).

Figure 1 — Owner–Member Set



DEPARTMENT (owner) connects to several EMPLOYEE records (members) through the set DEPT-EMPLOYEE.

Figure 2 — Many-to-Many via a Link Record



The link record ENROLLMENT is a member of two sets at once (STUDENT–ENROLLMENT and COURSE–ENROLLMENT) — this is what makes many-to-many relationships possible.

4. Characteristics

- Record-based organisation with named **set** relationships.
- A member record can have **multiple owners** → supports many-to-many relationships.
- **Navigational** (pointer/link-based) data access, not declarative querying.
- Relationships are **pre-defined** in the schema, giving fast, direct traversal.

5. Advantages vs Limitations

Advantages	Limitations
Fast, efficient navigation between linked records	Complex structure; difficult design
Represents complex and many-to-many relationships directly	Difficult schema maintenance
A record can participate in multiple relationships	Navigational access is complex to program
Reduces some redundancy vs. hierarchical model	Program–data dependence
Efficient, predictable traversal of connected data	Poor support for unplanned (ad-hoc) queries

■ IMPORTANT TERM — What is an ad-hoc query?

An **ad-hoc query** is a question you ask the database on the spot, without planning for it in advance — e.g. "show me all employees older than 40." In a network database, only the paths (sets) defined at design time can be followed, so unplanned questions like this are hard to answer.

6. Network vs Tree-Based Structure

Aspect	Network	Tree (Hierarchical)
Parent/owner	A member can have multiple owners	A child has exactly one parent
Relationships	1:M and, via link records, M:N	1:M only
Navigation	Multiple possible paths via sets	Single top-down path
Complexity	Higher	Lower

7. Quick Revision — 15 Facts

#	Fact
1	Network DB: records connected as a network, not a strict tree.
2	Model = record types + set types (owner–member relationships).
3	Set type: one owner, one or more member record types.
4	A member record can have multiple owners (key difference from a tree).
5	Set occurrence = one instance of a set type for a specific owner.
6	Links (pointers) physically connect owner and member records.
7	Many-to-many relationships need an intersection/link record (e.g., ENROLLMENT).
8	Charles Bachman (IDS, GE) — pioneer of the network model.
9	CODASYL's DBTG (1969/1971 report) formally defined the model.
10	Also known as the CODASYL model or DBTG model.
11	IDMS is a well-known commercial implementation.
12	Data access is navigational (FIND/GET-style), not declarative.
13	Main strength: efficient handling of complex, many-to-many relationships.
14	Main weakness: complex design and program-data dependence.
15	Largely succeeded by the relational model from the 1980s onward.

End of Chapter 4.1 — Network Database: Introduction (condensed edition).